

Heart Disease Prediction using Machine Learning

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Submission on: July 2026

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1. Project Overview

This Heart disease is one of the leading causes of mortality worldwide, making early diagnosis and risk assessment crucial for improving patient outcomes. This project analyzes patient health records age, blood pressure, cholesterol, ECG results, and related clinical attributes to predict heart disease risk, then packages that model into a usable prediction tool with risk categorization and health guidance.

2. Problem Statement

Cardiovascular disease is a leading cause of mortality worldwide, and early detection is critical for timely intervention. Conventional diagnosis relies on manual interpretation of clinical parameters — age, blood pressure, cholesterol, ECG, and chest pain type — a process that is often time-consuming and inconsistent.

This project develops a Machine Learning-based Heart Disease Prediction System using the UCI Heart Disease dataset. Through data preprocessing, model comparison, hyperparameter tuning, and deployment, the system predicts heart disease likelihood and delivers a probability score, risk category (Low/Medium/High), and health recommendations — usable by both patients and healthcare providers.

3. Objectives

- To collect and preprocess a heart disease dataset by handling missing values, removing duplicate records, treating outliers, and encoding categorical variables for model readiness.
- To perform exploratory data analysis (EDA) and develop multiple machine learning classification models, comparing their performance to identify the most effective baseline algorithm.
- To optimize the selected model through hyperparameter tuning and cross-validation, evaluating performance using industry-standard metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- To deploy the optimized model as an interactive web application that predicts heart disease risk, provides a probability score and risk category, and generates clinically meaningful health recommendations.

4.Dataset Description

4.1 Dataset Structure

The UCI Heart Disease dataset (Kaggle distribution) was used, containing patient records with 13 clinical attributes and 1 target label. Raw dataset size: 1,025 records x 14 columns.

Column	Description	Type
age	Age of the patient (years)	Numeric
sex	1 = Male, 0 = Female	Binary
cp	Chest pain type (0-3)	Categorical
trestbps	Resting blood pressure (mm Hg)	Numeric
chol	Serum cholesterol (mg/dl)	Numeric
fbs	Fasting blood sugar > 120 mg/dl (1=true)	Binary
restecg	Resting ECG results (0-2)	Categorical
thalach	Maximum heart rate achieved	Numeric
exang	Exercise induced angina (1=yes)	Binary
oldpeak	ST depression induced by exercise	Numeric
slope	Slope of peak exercise ST segment (0-2)	Categorical
ca	Number of major vessels colored by fluoroscopy (0-4)	Numeric/Ordinal
thal	Thalassemia (0-3)	Categorical
target	1 = Heart Disease present, 0 = Absent	Target

5.Tools and Technologies:

- **Programming Language:** Python
- **Libraries & Frameworks:**
 - **Pandas, NumPy** — data manipulation and preprocessing
 - **Matplotlib, Seaborn** — data visualization and EDA
 - **Scikit-learn** — model building, hyperparameter tuning (GridSearchCV), cross- validation, and evaluation metrics
 - **Joblib** — model serialization (saving/loading trained models)
 - **Streamlit** — web application deployment
- **Development Environment:**
 - Jupyter Notebook (Anaconda)
- **Version Control & Hosting:**
 - Git & GitHub

6.Methodology

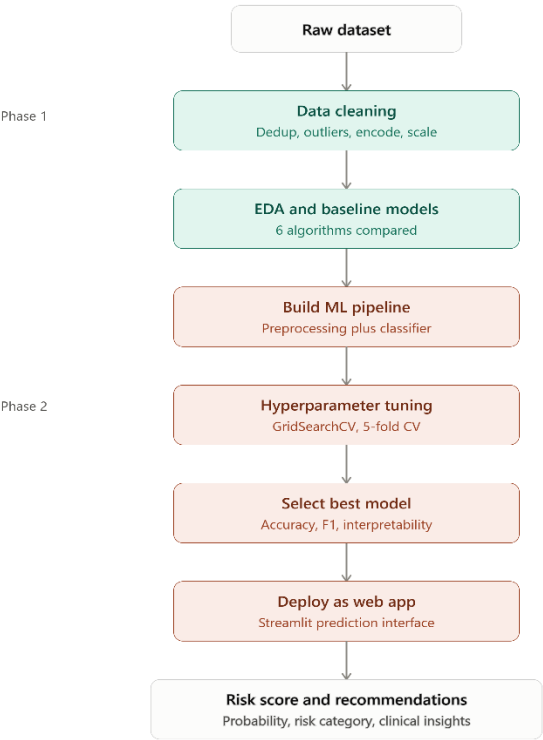
Phase 1: Data Preprocessing & Baseline Modelling

The UCI Heart Disease dataset (1,025 records) was collected and cleaned by removing 723 duplicate rows and verifying no missing values existed. Outliers in continuous features (trestbps, chol, thalach, oldpeak) were detected using the IQR method and capped. Nominal categorical variables (cp, restecg, slope, thal) were one-hot encoded, while continuous features were standardized using StandardScaler. Exploratory Data Analysis (EDA) was performed using correlation heatmaps, distribution plots, and boxplots to understand feature relationships. Six classification algorithms — Logistic Regression, Decision Tree, Random Forest, SVM, KNN, and Naive Bayes — were trained on an 80/20 train-test split and compared using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

Phase 2: Optimization, Deployment & Clinical Insights

A deployable scikit-learn Pipeline combining preprocessing and classification was constructed. Four candidate models were optimized using GridSearchCV with 5-fold Stratified Cross-Validation to identify optimal hyperparameters. Model performance was re-evaluated on the test set, and Logistic Regression was selected as the final model based on its balance of accuracy, precision, F1-score, and interpretability. The final pipeline was serialized using Joblib and deployed as an interactive Streamlit web application, allowing users to input patient parameters and receive a prediction probability, risk category (Low/Medium/High), and tailored health recommendations. Feature importance analysis of model coefficients was conducted to derive clinically meaningful insights into the key predictors of heart disease.

Workflow Diagram



PHASE 1 — Data Preprocessing, EDA & Baseline Modelling

3.1 Data Preprocessing

- Missing values: none found across all 14 columns.
- Duplicates: 723 exact duplicate rows removed (1,025 → 302 unique records).
- Outliers: detected via IQR method in trestbps, chol, thalach, oldpeak; capped (winsorized) rather than dropped to preserve sample size.
- Encoding: nominal categorical columns (cp, restecg, slope, thal) one-hot encoded; binary columns (sex, fbs, exang) and ordinal ca left unchanged.
- Scaling: continuous numeric features standardized (zero mean, unit variance) via StandardScaler.

3 Exploratory Data Analysis (EDA)

Fig 3.1: Target class distribution

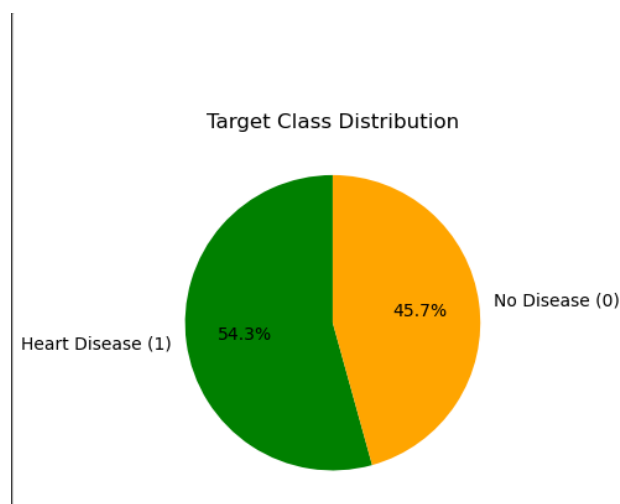


Fig 3.2: Correlation heatmap

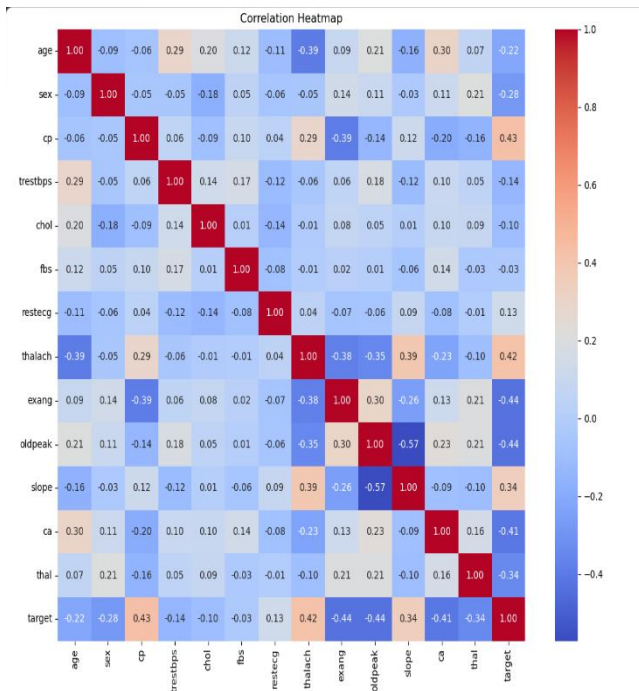


Fig 3.3: Chest pain

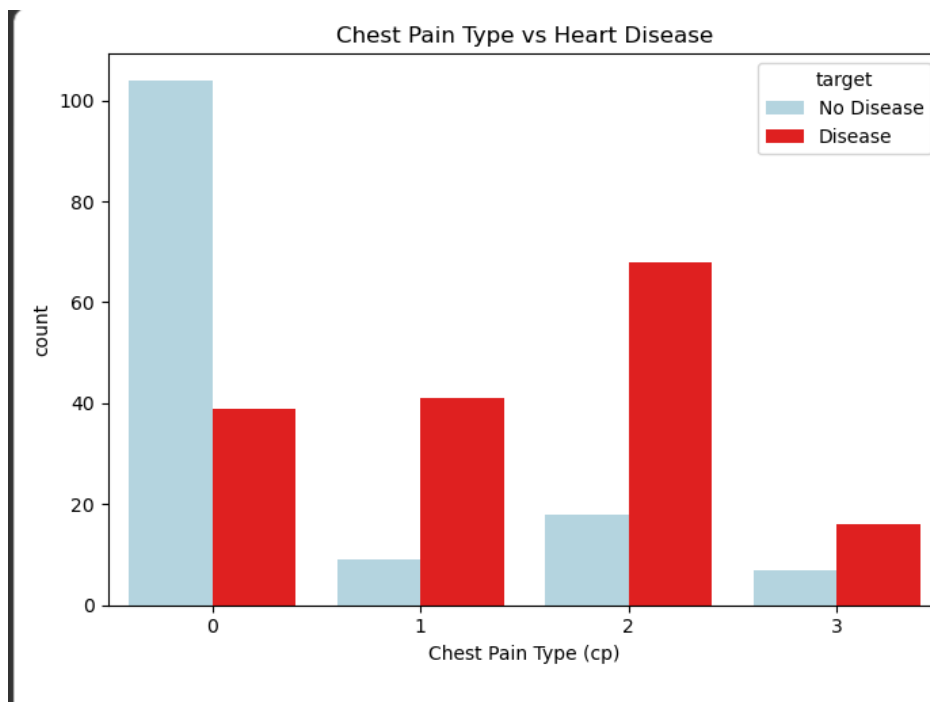


Fig 3.4: Boxplots of continuous features by target

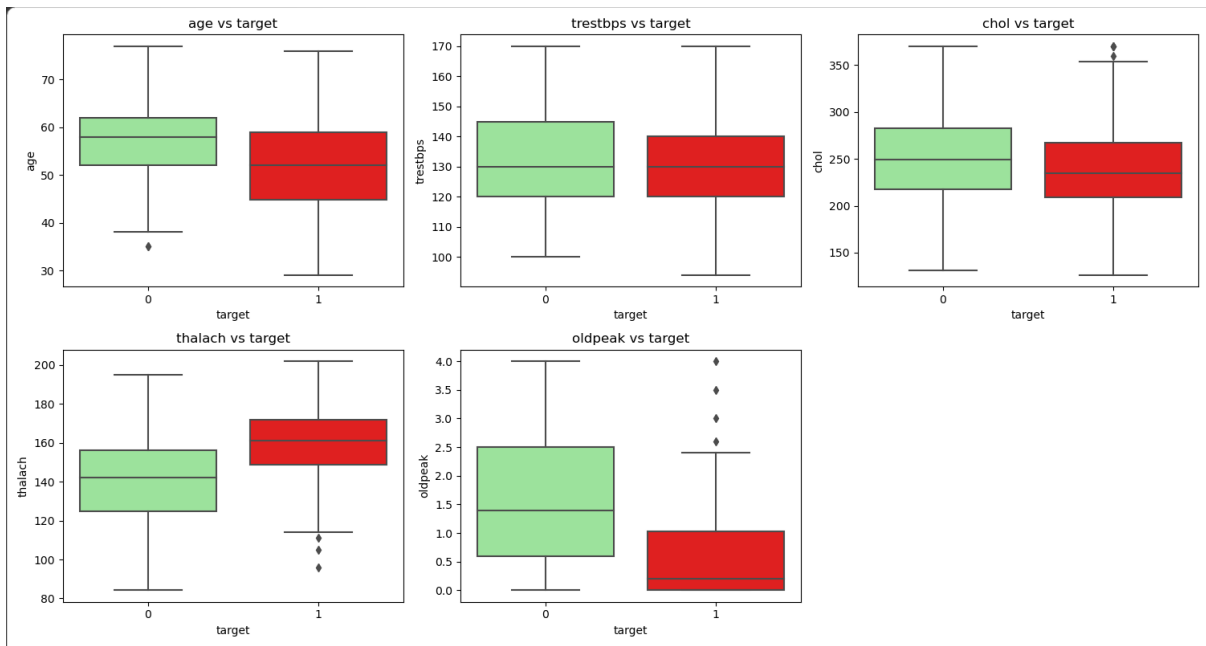


Table 3.5: Baseline Model Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.8525	0.8750	0.8485	0.8615	0.8983
Naive Bayes	0.8197	0.8438	0.8182	0.8308	0.8983
SVM (RBF)	0.8033	0.8182	0.8182	0.8182	0.8734
KNN	0.8033	0.8621	0.7576	0.8065	0.8864
Random Forest	0.7869	0.8125	0.7879	0.8000	0.8939
Decision Tree	0.7541	0.7500	0.8182	0.7826	0.7484

Fig 3.5.1: Model Comparison

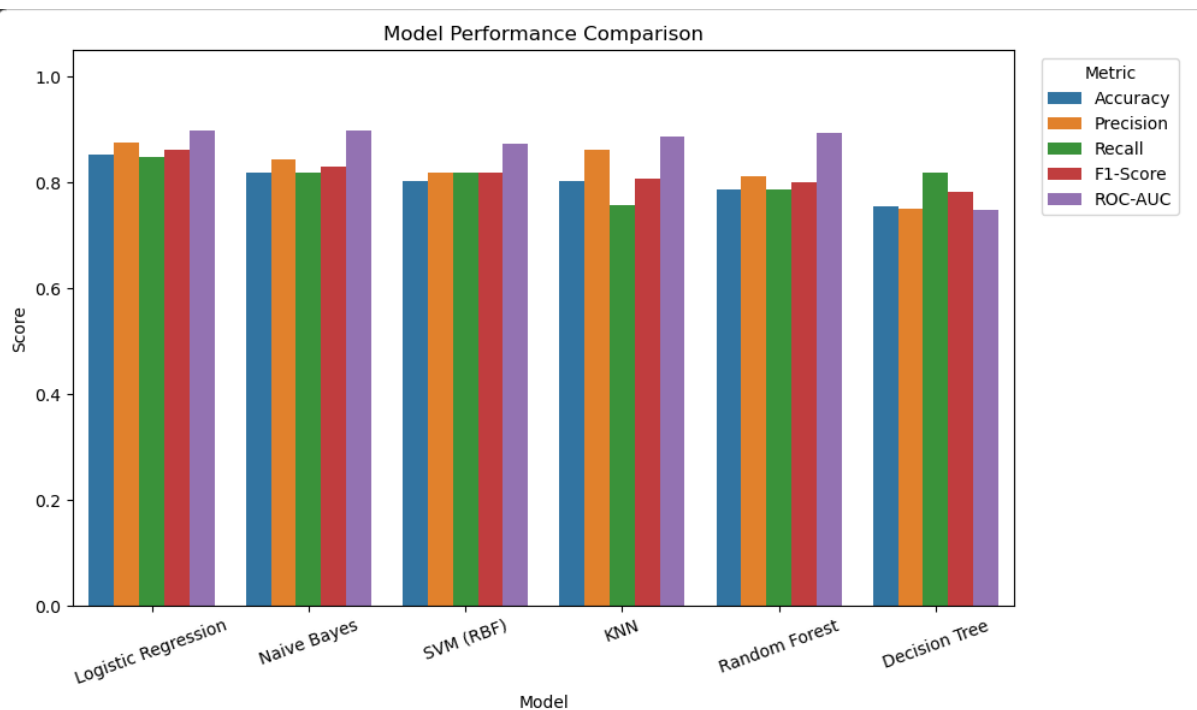
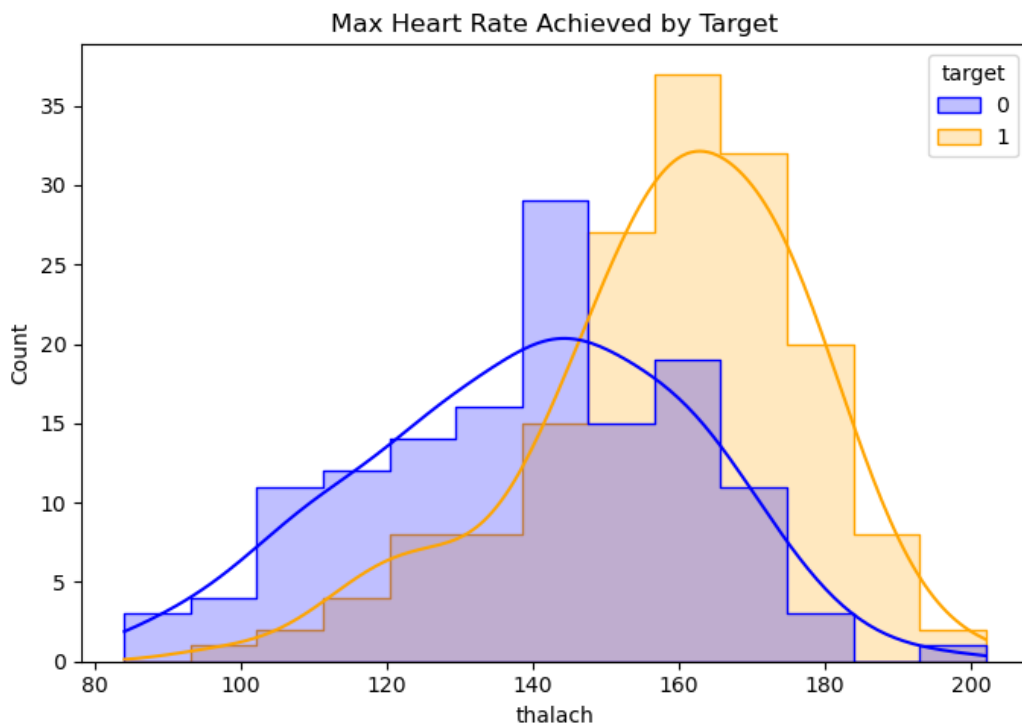


Fig 3.6: Max Heart Rate Achieved By Target



4.Key Findings

Logistic Regression was the top baseline performer (85.25% accuracy, 0.898 ROC-AUC), suggesting the relationship between clinical features and heart disease risk is largely linear after preprocessing.

PHASE 2 — Model Optimization, Deployment & Clinical Insights

4.1 Hyperparameter Tuning & Cross-Validation

To move beyond default hyperparameters, a single deployable scikit-learn Pipeline (ColumnTransformer preprocessing + classifier) was built for each of four candidate models: Logistic Regression, Random Forest, SVM (RBF), and KNN. Each was tuned via GridSearchCV with 5-fold Stratified Cross-Validation, optimizing for ROC-AUC.

- Logistic Regression — best params: C=1, solver=liblinear, penalty=l2
- Random Forest — best params: n_estimators=100, max_depth=7, min_samples_split=5, min_samples_leaf=1
- SVM (RBF) — best params: C=10, gamma=0.01
- KNN — best params: n_neighbors=11, weights=distance, p=1

Table 4.2: Tuned Model Comparison

Model	Accuracy	Test Accuracy	Test Precision	Test Recall	Test F1-Score	Test ROC-AUC
SVM (RBF)	0.8463	0.8361	0.8485	0.8485	0.8485	0.9091
Naive Bayes	0.8088	0.8197	0.8438	0.8182	0.8308	0.8983
KNN	0.8131	0.8361	0.8710	0.8182	0.8438	0.8961
Logistic Regression (Deployed) ✓	0.8421	0.8525	0.8750	0.8485	0.8615	0.8939
Random Forest	0.8213	0.7869	0.8125	0.7879	0.8000	0.8799
Decision Tree	0.7761	0.7541	0.7813	0.7576	0.7692	0.8274

Fig 4.2.1: Post-tuning performance comparison across all metrics.



Fig 4.2.2: ROC curves for all tuned models

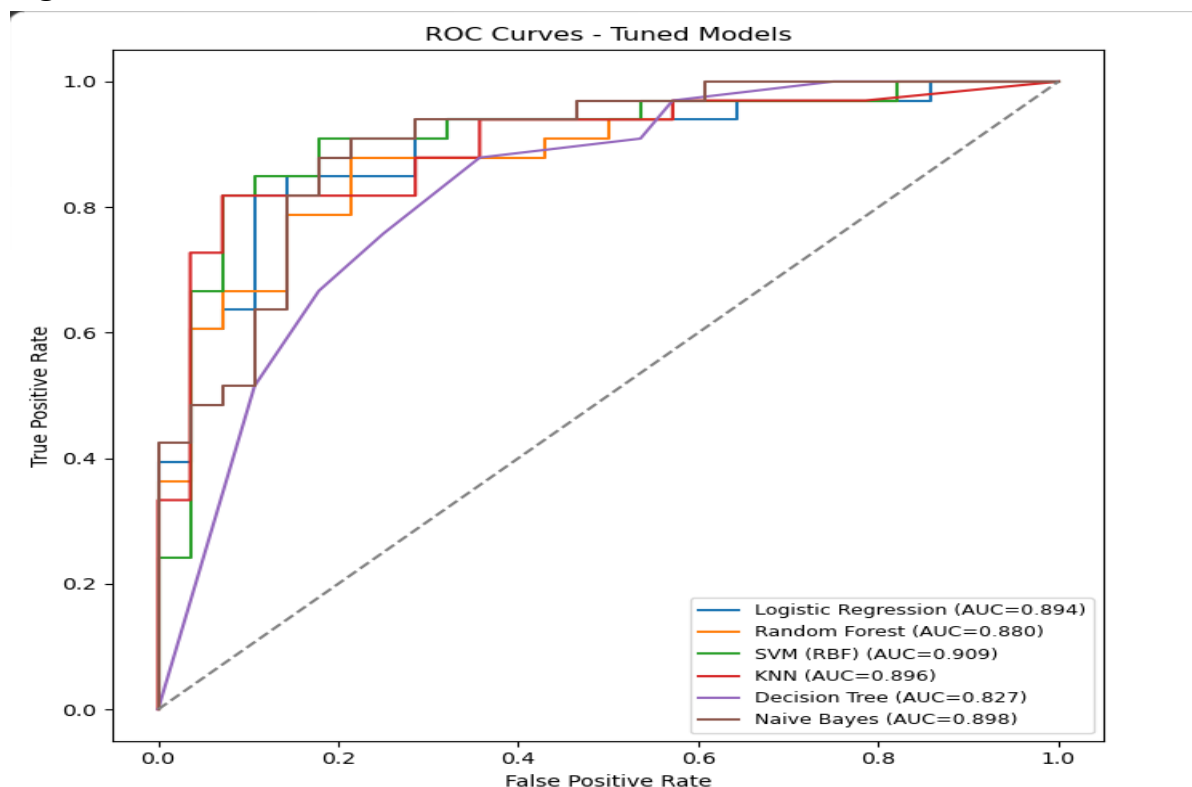


Fig 4.3: Final Model Selection

SVM (RBF) achieved the single highest ROC-AUC (0.909), but the margin over Logistic Regression (0.894) is marginal, and Logistic Regression outperforms it on Accuracy (85.25% vs 83.61%), Precision (0.875 vs 0.849), and F1-Score (0.862 vs 0.849). Critically, Logistic Regression is fully interpretable via its coefficients, while SVM with an RBF kernel is effectively a black box. Since this project requires generating clinical insights and health recommendations directly from the model, interpretability is a deployment requirement — not a nice-to-have. Logistic Regression was therefore selected as the final deployed model.

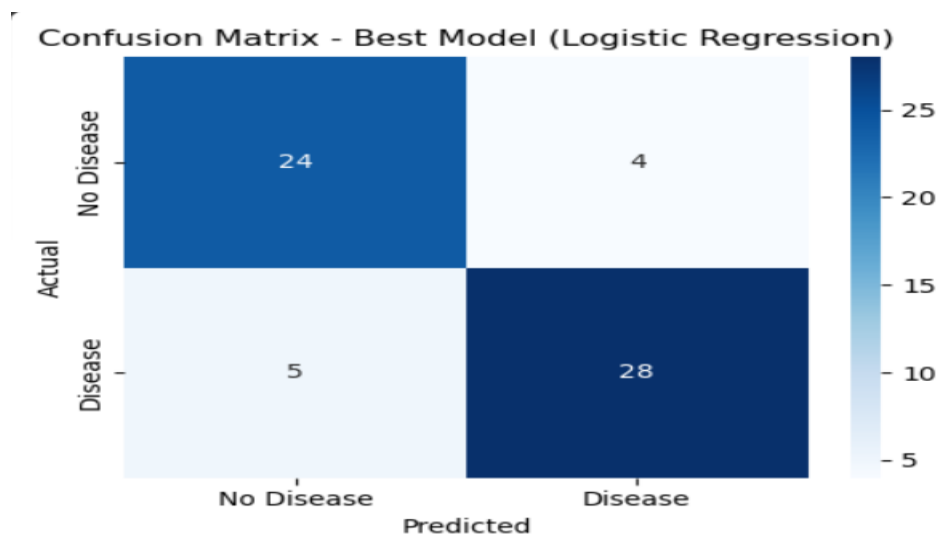


Fig 4.4: Feature Importance Analysis

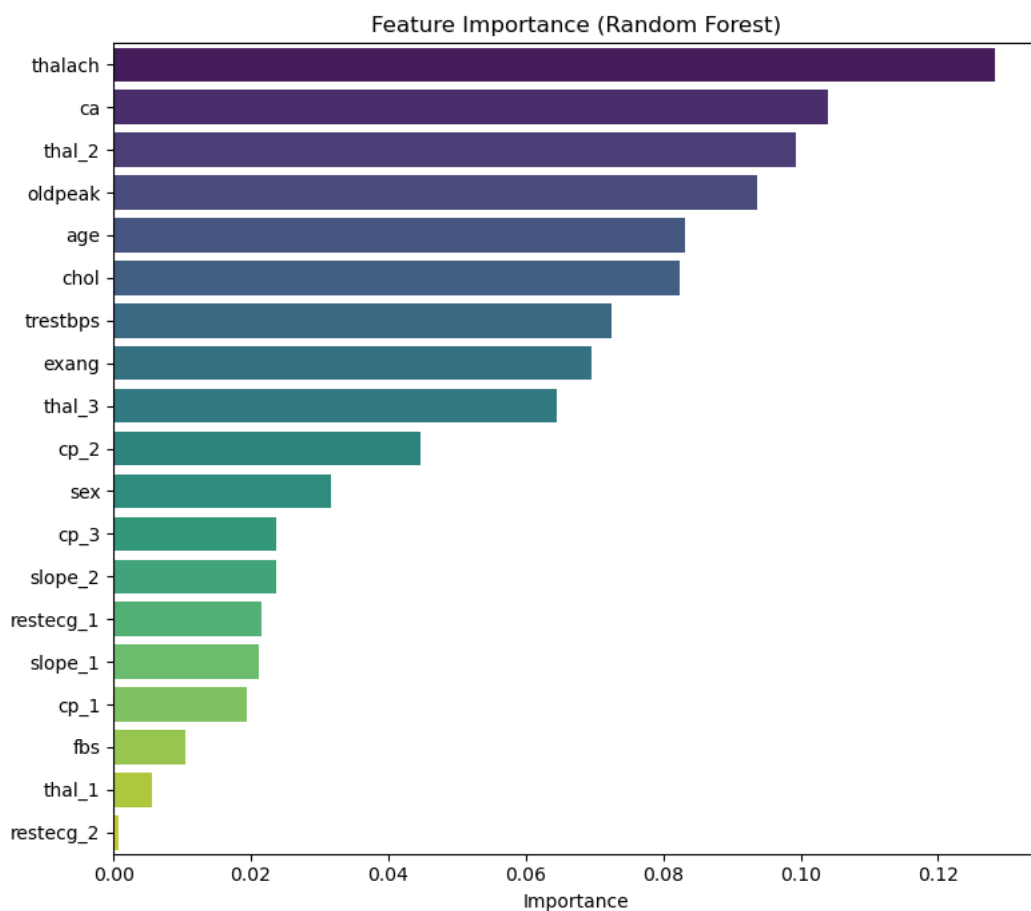
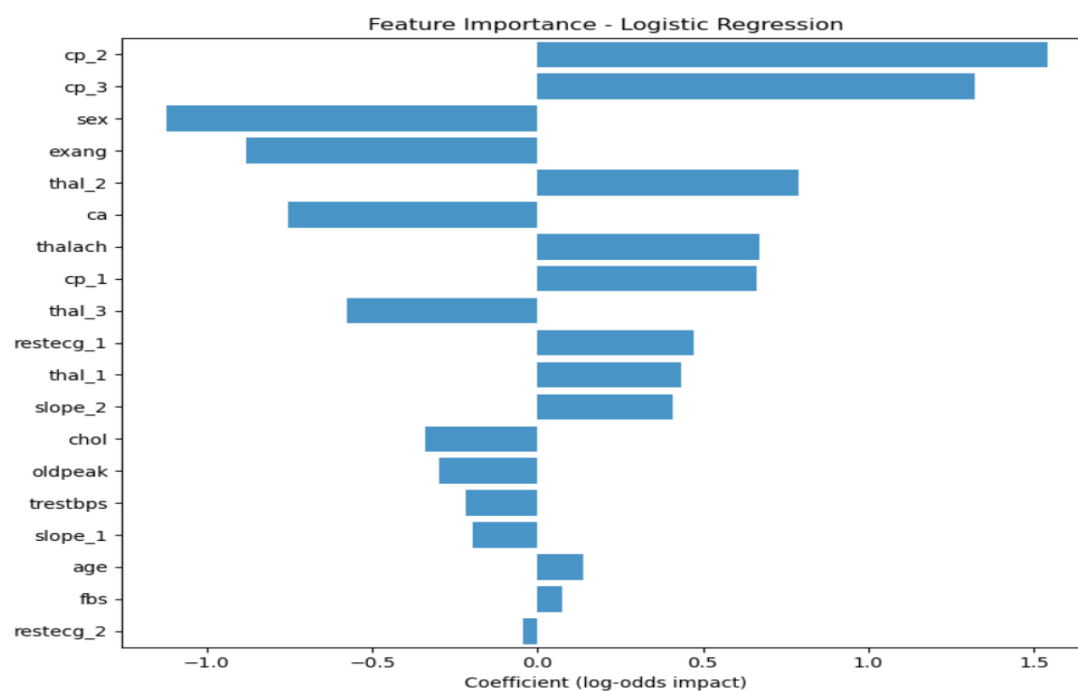
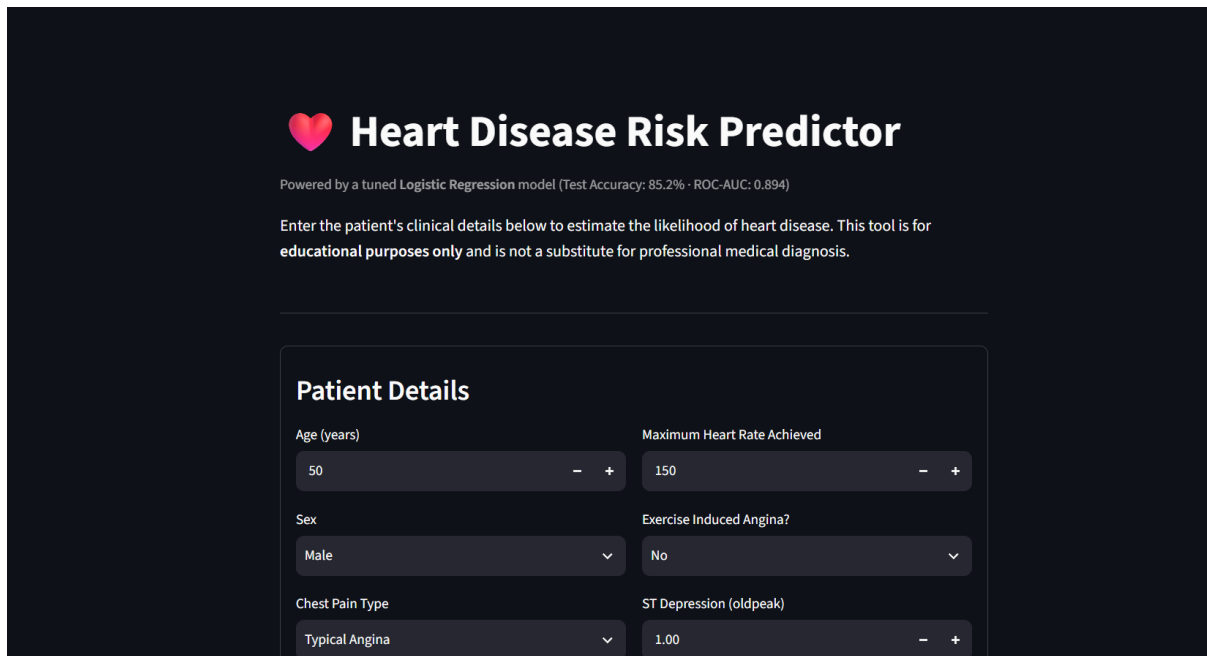


Fig 4.4.1: Logistic Regression coefficients



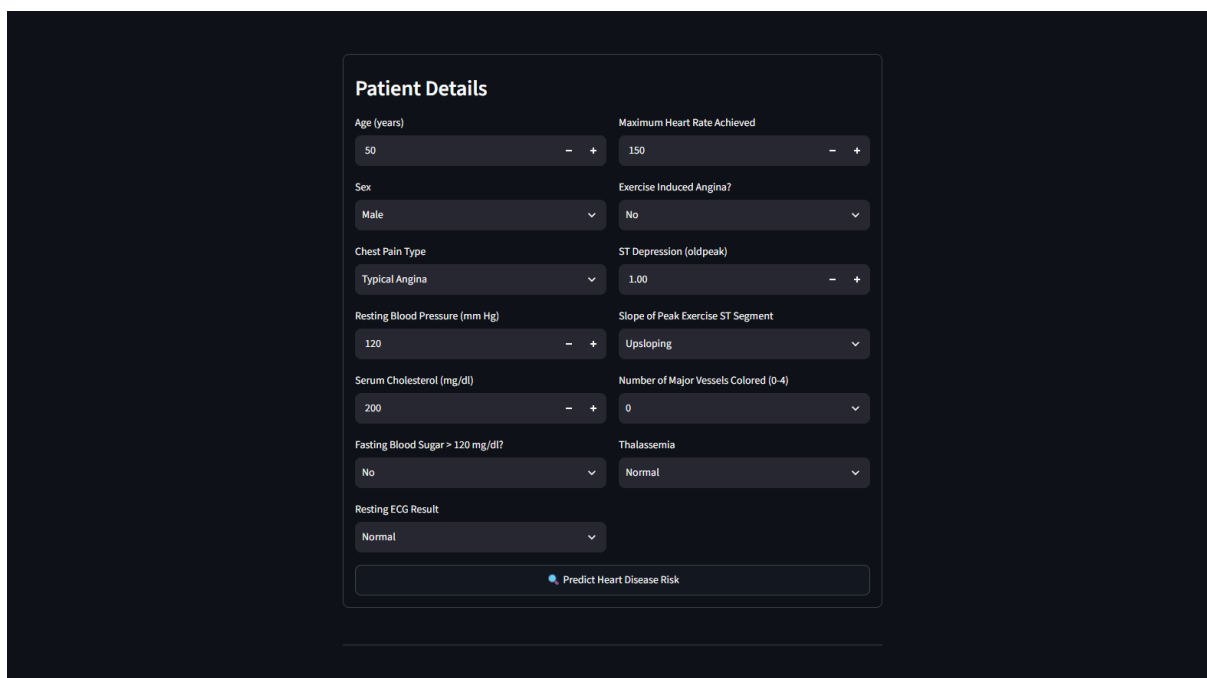
5. Deployment

The final model was packaged as a single deployable scikit-learn Pipeline (heart_disease_best_pipeline.pkl) that accepts raw clinical inputs directly — handling scaling and encoding internally — and deployed as an interactive Streamlit web application (app/app.py).



The screenshot shows the 'Heart Disease Risk Predictor' web application. At the top, there is a red heart icon and the title 'Heart Disease Risk Predictor'. Below the title, a subtitle states: 'Powered by a tuned Logistic Regression model (Test Accuracy: 85.2% · ROC-AUC: 0.894)'. A disclaimer follows: 'Enter the patient's clinical details below to estimate the likelihood of heart disease. This tool is for educational purposes only and is not a substitute for professional medical diagnosis.' The main form is titled 'Patient Details' and contains several input fields: 'Age (years)' with a value of 50, 'Maximum Heart Rate Achieved' with a value of 150, 'Sex' with a dropdown menu set to 'Male', 'Exercise Induced Angina?' with a dropdown menu set to 'No', 'Chest Pain Type' with a dropdown menu set to 'Typical Angina', and 'ST Depression (oldpeak)' with a value of 1.00. The form is styled with a dark background and light-colored text.

Fig 5.1: User Interface



This screenshot shows the same 'Heart Disease Risk Predictor' web application, but with more input fields visible. The 'Patient Details' form includes: 'Age (years)' (50), 'Maximum Heart Rate Achieved' (150), 'Sex' (Male), 'Exercise Induced Angina?' (No), 'Chest Pain Type' (Typical Angina), 'ST Depression (oldpeak)' (1.00), 'Resting Blood Pressure (mm Hg)' (120), 'Slope of Peak Exercise ST Segment' (Upsloping), 'Serum Cholesterol (mg/dl)' (200), 'Number of Major Vessels Colored (0-4)' (0), 'Fasting Blood Sugar > 120 mg/dl?' (No), 'Thalassemia' (Normal), and 'Resting ECG Result' (Normal). At the bottom of the form is a button labeled 'Predict Heart Disease Risk' with a small blue icon.

Fig 5.2 User Input

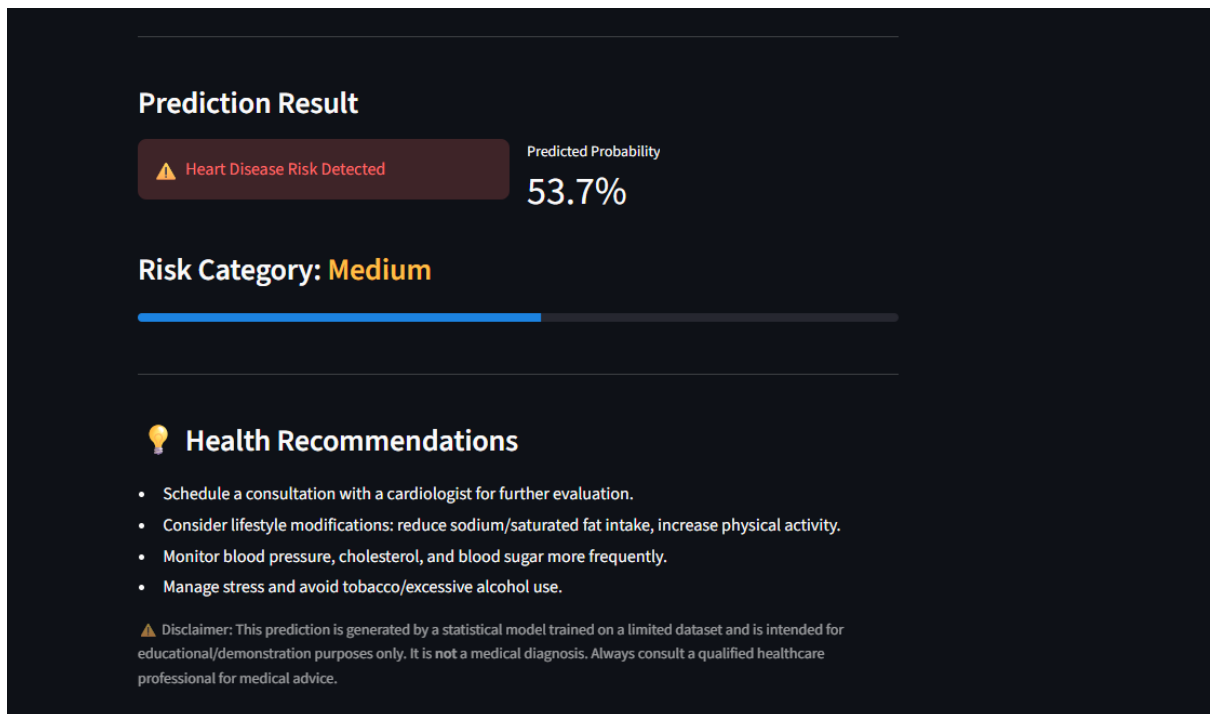


Fig 5.3: Model Prediction Based on User Input

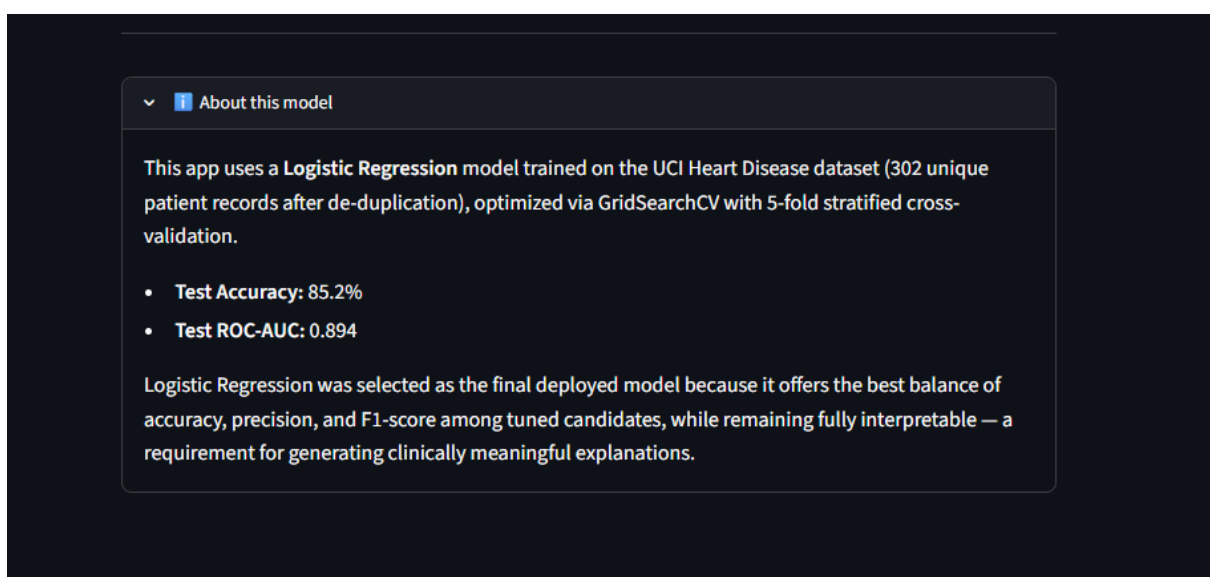


Fig 5.4: Model Rating

5.1 Application Features

- Structured input form for all 13 clinical parameters (age, sex, chest pain type, blood pressure, cholesterol, ECG results, etc.)
- Instant prediction: Heart Disease Risk Detected / No Significant Risk Detected
- Prediction probability displayed as a percentage and progress bar
- Risk category classification: Low (<33%), Medium (33-66%), High (>66%)
- Tailored health recommendations generated per risk category
- Model transparency panel showing accuracy, ROC-AUC, and model choice rationale
- Medical disclaimer clarifying the tool is for educational/demonstration purposes only

To run: install requirements.txt, then execute 'streamlit run app/app.py' from the app directory

6. Clinical Insights

Based on the Logistic Regression coefficients (Figure 9), the following clinically-relevant patterns emerged from this dataset:

- Chest pain type (cp): non-typical chest pain (atypical angina, non-anginal pain, asymptomatic) is associated with substantially higher predicted risk than typical angina — the strongest predictor in the model.
- Thalassemia (thal): fixed and reversible defects shift predicted risk meaningfully relative to normal thalassemia results.
- Maximum heart rate (thalach): higher achieved heart rate during testing is associated with higher predicted risk in this dataset.
- Number of major vessels (ca): a higher count is associated with lower predicted risk in this particular encoding — a reminder to always verify label/feature directionality against the dataset's documentation before drawing conclusions.
- ST depression (oldpeak) and cholesterol (chol): both show a negative association with predicted risk in this model, which runs counter to some general clinical intuition and illustrates why multivariate model coefficients should not be read as isolated univariate medical facts.

These are statistical associations learned from a relatively small dataset (302 records) and reflect patterns in this specific sample. They should not be interpreted as established medical causation. Any real-world clinical application would require validation on larger, prospectively collected, and demographically diverse cohorts, together with review by qualified medical professionals

7. Conclusion

This project successfully delivered an end-to-end machine learning pipeline for heart disease risk prediction — from raw data cleaning through to a deployed, interactive prediction tool. The tuned Logistic Regression model achieves strong, well-balanced performance (85.25% accuracy, 0.894 ROC-AUC) while remaining fully interpretable, enabling the system to pair each prediction with a probability score, a risk category, and clinically grounded health recommendations.

8. Future Work

- Validate on a larger, more diverse patient dataset to improve generalizability.
- Add SHAP-based per-patient explanations to the app for individualized clinical reasoning.
- Extend deployment with authentication and persistent prediction history for clinical use cases.
- Explore ensemble/stacking approaches combining Logistic Regression with SVM for a possible accuracy/interpretability trade-off.